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Post-Licensure Herpes Zoster Vaccines Effectiveness: Systematic Review Protocol

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Post-Licensure Herpes Zoster Vaccines Effectiveness: Systematic Review Protocol.

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ABSTRACT

Introduction

Herpes zoster (HZ) and associated complications inflict substantial morbidity, and associated healthcare and socioeconomic burdens. Current treatments are not fully effective, especially among the most vulnerable populations. Two HZ vaccines are available and are part of the national immunisation programmes in many countries. This review will evaluate the effectiveness of zoster vaccines against incident herpes zoster and postherpetic neuralgia in adults 50 years and older.

Methods and analysis

The key information sources that will be searched include Medline (Ovid), Embase (Ovid), Cochrane libraries, and CINAHL. This search will consider post-licensure observational studies published in all languages between 2006 and 2020 that assessed the effectiveness of herpes zoster/zoster vaccines in adults 50 years and older. The identification of studies will be complemented with the search of reference lists and citations, and contact with authors of papers to request missing or additional data, where required. Following the search, all identified citations will be collated, and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review. Selected studies will follow the process of critical appraisal, data extraction, and data synthesis. Statistical analyses will be performed using a random-effects model.

Ethics and dissemination

Formal ethical approval is not required, as primary data will not be collected. The review will be disseminated in peer-reviewed publications and conference presentations.

Keywords: Herpes zoster vaccine, Effectiveness

Strengths and limitations of this study

- Systematic review of high-quality observational studies.
- The selection of studies, data extraction, and a quality assessment will be conducted by two independent authors.
- The protocol has been created according to published PRISMA-P guidelines.
- We do not have any language limitations.

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- Observational studies may produce evidence of possible bias.

INTRODUCTION

Varicella-zoster virus (VZV) is the aetiological agent of varicella, a highly infectious, acute, self-limiting, viral disease with serious complications in neonates, pregnant women, and immunocompromised persons (1). It is an exclusively human neurotropic alpha herpes virus (2). After primary infection, it becomes latent in the cranial nerve ganglia, dorsal root ganglia, and autonomic ganglia along the entire neuroaxis (2). Administration of varicella vaccine or natural infection generates VZV-specific antibody and VZV-specific T cell-mediated immunity (2).

The memory cell response contributes to the protection following re-exposure to VZV (3). This immunity is subsequently boosted either by endogenous re-exposure (subclinical reactivation of latent VZV) or exogenous re-exposure (3). VZV-specific T cell-mediated immunity is necessary to maintain latent VZV in a subclinical state in the sensory ganglia (4). A decline in VZV-specific cell-mediated T cell-mediated immunity associated with ageing (immunosenescence), immune-suppressive diseases or therapies causes reactivation of latent VZV as herpes zoster (HZ) (4) .

HZ presents as a unilateral vesicular rash, characteristically restricted to a single dermatome, and accompanied by radicular pain (5). About one million cases of HZ are reported every year (6). HZ incidence ranges from 3·4 to 5·0 per 1000 person-years, and 8 to 11 per 1000 person-years over the age of 65. One in four people is at risk of developing HZ during their lifetime, and two-thirds of people with the disease are aged 50 years or older (7). By the age of 85 years, >50% of the population report at least one episode of HZ (2). Major risk factors for HZ are increasing age, and a decline in cell-mediated immunity (5).

Laboratory confirmation of HZ occurs through the detection of VZV DNA using polymerase chain reaction or isolating VZV in cell culture from vesicular fluid, crusts, saliva, cerebrospinal fluid, or other specimens (5).

The main complications of HZ include post-herpetic neuralgia (PHN - pain persisting more than 90 days after rash onset) in 22% (8-26%) of HZ patients; herpes zoster ophthalmicus

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(HZO), ophthalmic division of trigeminal nerve ~15% of cases (if untreated, 50-70% develop acute ocular complications, chronic complications, reduced vision, and blindness) (5).

The immune response can be boosted by the VZV vaccine developed to prevent HZ. Vaccination is, therefore, an important tool to reduce the epidemiological, clinical and economic burden of HZ, and to reduce the negative impact on the quality of life (2). Currently, two HZ vaccines are available. The recombinant zoster vaccine (RZV), approved in 2017, and the Zoster Vaccine Live (ZVL) licensed in 2006 (8). These HZ vaccines are effective, safe, and generally well tolerated. RZV is also known as adjuvanted recombinant subunit zoster vaccine and consists of recombinant VZV glycoprotein E, and a liposome-based AS01B adjuvant system (8). These vaccines have been included in the national immunisation programs of many countries (1).

HZ and associated complications inflict substantial morbidity, and associated healthcare, and socioeconomic burdens (9). In the US, HZ and associated diseases result in about $1·3 billion in medical care costs and $1·7 billion in indirect costs annually, and this burden will increase in the coming years due to the ageing populations (9). Current treatments are not fully effective, especially among the most vulnerable patients (10). The best available option is prevention through vaccination. These vaccines have been licensed and marketed in many high-income countries (1).

A 2019 Cochrane systematic review of 24 studies (randomised controlled trials or quasi-randomised controlled trials) involving 88,531 participants, aged 60 years and above showed that ZVL and RZV are effective in preventing herpes zoster disease for up to three years (11). This review excluded (a) trials involving immunosuppressed persons, (b) trials in people <60 years, and (c) observation studies.

In 2018, Tricco et al compared the efficacy, effectiveness, and safety of the herpes zoster live attenuated vaccine with the herpes zoster adjuvant recombinant subunit vaccine or placebo for adults aged 50 and older. They concluded that the RZV might prevent more cases of HZ than using the ZVL, but the RZV also carries a greater risk of adverse events at injection sites (7). Of all the 27 studies included, there were only four observational studies (1 case-control, and 3 cohorts). For the laboratory or doctor confirmed cases of HZ, ZVL was not statistically different from placebo. There is a need to further explore these findings in observational studies.

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Another systematic review estimated the relative efficacy and safety of vaccines for the prevention of HZ using network meta-analysis. RZV was significantly more effective in reducing HZ and PHN incidence in adults 60 years, and above, compared with ZVL. RZV resulted in more reactogenicity following immunization (12). The review excluded (a) trials conducted in immunocompromised patients, (b) trials in participants less than 50 years, and (c) observational studies.

In 2019, Senderovich et al, summarized the current literature available on the efficacy of the HZ vaccines in adults over 60 years old, and evaluated the cost-effectiveness of the HZ vaccines. 10 studies that met the inclusion criteria highlighted the efficacy of the HZ vaccines (6). The review focused on adults over 60 years old residing in long-term care facilities, and was limited to a five-year period (January 2013 to April 2018).

There is a need for a systematic review that will summarise real-world evidence (13) of the effectiveness of herpes zoster vaccines from post-licensure studies with different designs, conducted in different populations. Vaccine effectiveness (protection attributable to a vaccine administered under field conditions to a given population) measured by observational post-licensure studies (14) is important to inform public health programs and policies. The evidence will be obtained by searching, critically appraising, and synthesising the evidence from observational studies of published, and unpublished literature.

This present study will be the only review that will evaluate, and summarise the post-licensure effectiveness of HZ vaccines from 2006 to 2020 in adults 50 years and above.

A preliminary search of PROSPERO, MEDLINE, the Cochrane Database of Systematic Reviews and the *JBI Database of Systematic Reviews and Implementation Reports* was conducted and no current or underway systematic reviews on the topic were identified.

OBJECTIVES

We seek to evaluate the effectiveness of zoster vaccines in adults 50 years and older. This includes incidence of herpes zoster, postherpetic neuralgia and herpes zoster ophthalmicus (primary outcomes), and herpes zoster related hospitalisations and quality of life (secondary outcomes)).

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METHODS

The proposed systematic review will be conducted in accordance with the Joanna Briggs Institute methodology for systematic reviews of effectiveness evidence (15, 16).

Design

We will conduct a systematic review of published and unpublished studies.

Eligibility criteria

Types of studies: This review will consider analytical observational studies including prospective and retrospective cohort studies, and case-control studies. Studies published in all languages will be included. Studies published from 2006 to present will be included as the first herpes zoster vaccine was licensed in 2006. Systematic reviews and meta-analyses will be used only for identifying additional studies; these will not be included in this review.

Types of participants: The review will consider studies that include adults aged 50 years and older (studies can be included when adults younger than 50 are included, so long as this is not the majority).

Types of interventions: This review will consider studies that evaluate the effectiveness of herpes zoster vaccines. The invention is being vaccinated (live attenuated injectable herpes zoster vaccine, adjuvant recombinant subunit herpes zoster vaccine). i.e. any licensed HZ vaccine (including all preparations, schedules, dosing and routes of administration).

Types of comparators: This review will consider studies that compare the intervention to no vaccination, placebo, no vaccine or other vaccine

Types of outcome measures: This review will consider studies that include the following outcomes: The outcome measures are defined in Appendix III

Primary outcomes

1. Incidence of herpes zoster (entire duration of follow-up or repeated measures).
   - Number of cases and person-years
   - Incidence rate per arm

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2. Incidence of Postherpetic neuralgia (entire duration of follow-up)
   - 30 days, 60 days, 90 days
   - Number of cases and person-years
   - Incidence rate per arm
3. Incidence of other HZ complications (herpes zoster ophthalmicus)
   - Number/proportion of events or number/proportion of patients with the event
   - Incidence rate

Secondary outcomes

- Herpes zoster related hospitalisations
- Quality of life

These outcomes will be measured by vaccine effectiveness.

Types of studies

This review will consider analytical observational studies including prospective and retrospective cohort studies, and case-control studies. Studies published in all languages will be included. Studies published from 2006 to present will be included as the first herpes zoster vaccine was licensed in 2006.

Search Methods

The search strategy will aim to locate both published and unpublished studies. An initial limited search of MEDLINE was undertaken to identify articles on the topic. The text words contained in the titles and abstracts of relevant articles, and the index terms used to describe the articles were used to develop a full search strategy for Medline (Ovid) (see Appendix I). The search strategy, including all identified keywords and index terms, will be adapted for each included information source. The reference list of all studies selected for critical appraisal will be screened for additional studies.

The databases to be searched include Medline (Ovid), Embase (Ovid), Cochrane libraries, Cumulative Index to Nursing and Allied Health Literature (CINAHL), ProQuest Central, and Dimensions. Sources of unpublished studies and grey literature to be searched include CareSearch, Grey Literature Report, Google, and WHO.

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Study selection

Following the search, all identified citations will be collated and uploaded into EndNote X9 (Clarivate Analytics, PA, USA), and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review. Potentially relevant studies will be retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI) (Joanna Briggs Institute, Adelaide, Australia) (16).

The full text of selected citations will be assessed in detail against the inclusion criteria by two independent reviewers. Reasons for exclusion of full-text studies that do not meet the inclusion criteria will be recorded and reported in the systematic review. Any disagreements that arise between the reviewers at each stage of the study selection process will be resolved through discussion, or with a third reviewer. The results of the search will be reported in full in the final systematic review and presented in Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) flow diagram (17).

Assessment of methodological quality

Eligible studies will be critically appraised by two independent reviewers at the study level for methodological quality in the review using standardized critical appraisal instruments from the Joanna Briggs Institute for experimental and quasi-experimental studies (15, 16). Authors of papers will be contacted to request missing or additional data for clarification, where required. Any disagreements that arise will be resolved through discussion, or with a third reviewer. The results of the critical appraisal will be reported in narrative form and in a table.

All studies, regardless of the results of their methodological quality, will undergo data extraction and synthesis (where possible).

Data extraction

Data will be extracted from studies included in the review by two independent reviewers using the standardised data extraction tool.

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The data extracted will include specific details about the populations, study methods, interventions, and outcomes of significance to the review objective (HZ, HZ ophthalmicus, postherpetic neuralgia, HZ related hospitalisations, and quality of life).

Any disagreements that arise between the reviewers will be resolved through discussion, or with a third reviewer. Authors of papers will be contacted to request missing or additional data, where required.

Data analysis

Studies will, where possible be pooled in a statistical meta-analysis using JBI SUMARI (16). Effect sizes will be expressed as either odds ratios or relative risk (for dichotomous data) and weighted (or standardized) final post-intervention mean differences (for continuous data) and their 95% confidence intervals will be calculated.

Heterogeneity will be assessed statistically using the standard chi-squared and I squared tests. Statistical analyses will be performed using a random-effects model (18).

Sensitivity analyses will be conducted to test decisions made regarding the impact of excluding or including studies in a meta-analysis based on sample size, or methodological quality. If results remain consistent across the different analyses, the results can be considered robust as even with different decisions they remain the same/similar. If the results differ across sensitivity analyses, this is an indication that the result may need to be interpreted with caution (15).

Where statistical pooling is not possible the findings will be presented in narrative form including tables and figures to aid in data presentation where appropriate. A funnel plot will be generated to assess publication bias if there are 10 or more studies included in a meta-analysis. Statistical tests for funnel plot asymmetry (Egger test, Begg test, Harbord test) will be performed where appropriate.

Assessing certainty in the findings

The Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach for grading the certainty of evidence will be followed (15, 19), and a Summary of Findings (SoF) will be created using GRADEPro GDT (McMaster University, ON, Canada).

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The SoF will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence-based on the risk of bias, directness, heterogeneity, precision and risk of publication bias of the review results. The outcomes reported in the SoF will be herpes zoster, herpes zoster ophthalmicus, postherpetic neuralgia, herpes zoster related hospitalisations and quality of life.

Patient and Public Involvement statement

No patient involved

Ethics and dissemination

Formal ethical approval is not required, as primary data will not be collected. The findings will be disseminated in peer-reviewed journals and conference presentations.

Conclusion

This systematic review will provide evidence regarding the effectiveness of herpes zoster vaccination in people 50 years of age and above. The evidence will be obtained by searching, critically appraising, and synthesising the evidence from observational studies of published and unpublished literature.

Registration and publishing statement

The systematic review protocol will be registered with the PROSPERO International Prospective Register of Systematic Reviews, and reported using Preferred Reporting Items for Systematic Reviews and Meta-analysis checklist to guide the reporting of the review.

Contributors

MFJ wrote this review. CRS and BPN contributed to conceiving this review, defining the study question and planning the review, and commented critically on several drafts of the protocol. PAM and PEE contributed on a draft of this review with critical comments.

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Conflicts of interest

Professor Simpson reports grants from the UK National Institute for Health Research, UK Medical Research Council, the NZ Health Research Council and Ministry of Business Innovation and Employment. Dr. Nguyen reports grants from the NZ Health Research Council and NZ Ministry of Business Innovation and Employment.

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APPENDICES

Appendix I: Search strategy

Key concepts (alternative words)

1. Herpes zoster (zoster, shingles, varicella-zoster virus, human herpesvirus 3)

2. Postherpetic neuralgia (PHN)

3. Herpes zoster ophthalmicus (HZO)

4. Herpes zoster vaccine (zoster vaccine, shingles vaccine, zostavax, live attenuated zoster vaccine, zoster vaccine live (ZVL), shingrix, adjuvant recombinant subunit vaccine (RZV), varicella-zoster vaccine).

5. Effectiveness (effective, performance, impact, efficacy, effect, efficiency).

6. Other search terms to be considered: HZ related hospitalisations, quality of life, immune status, comorbidities.

Search conducted on Medline (Ovid) show the following

#	Search term	Results
1	exp Herpes Zoster/	11747
2	Herpesvirus 3, Human/	7028
3	Varicellovirus/	285
4	shingles.tw.	1240
5	(varicella adj3 virus*).tw.	8299
6	zoster.tw.	17343
7	(hhv3 or hhv-3).tw.	16
8	shingles.ab,ti.	1240
9	zoster.ab,ti.	17343
10	(varicella adj4 virus*).ab,ti.	8370
11	varicellovirus/de	27
12	varicellovir*.ab,ti.	62
13	hhv3.ab,ti.	5
14	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13	22283
15	exp Vaccines/	225746
16	exp Immunization/	173130
17	Vaccination/	80276
18	(vaccin* or immuni* or inocul*).tw.	636177
19	(vaccin* or immuni* or inocul*).ab,ti.	636177
20	15 or 16 or 17 or 18 or 19	724546
21	14 and 21	4827
22	Herpes Zoster Vaccine/	790
23	((zoster or shingles) adj3 vaccin*).tw.	1264
24	zostavax.tw,nm.	132
25	((sub-unit or subunit) adj3 (vaccin* or zoster)).ab,ti.	5893

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26	(Zostavax or Shingrix).ab,ti.	153
27	22 or 23 or 24 or 25 or 26	7342
28	21 or 27	10650
29	exp cohort analysis/	1978236
30	exp longitudinal study/	132972
31	exp prospective study/	534668
32	cohort*.tw.	573867
33	exp case control study/	1069768
34	(case* and control*).tw.	688836
35	29 or 30 or 31 or 32 or 33 or 34	2751394
36	28 and 35	989
37	limit 36 to yr="2006 -Current"	696
38	remove duplicates from 37	692

Appendix II: Data extraction instrument

Standardised JBI data extraction instrument.

Appendix III: Outcome Definitions

1. Confirmed herpes zoster infection: Those who underwent clinical diagnosis of herpes zoster infection through laboratory testing (e.g. polymerase-chain-reaction assay, virus culture) and/or examination by a physician were classified as having confirmed cases of herpes zoster infection (1).

2. Suspected herpes zoster infection: Subjects with non-injection-site herpes-zoster like rashes, varicella-like rashes, vesicular rashes or unilateral rashes (as defined by the investigators) and no other alternative clinical diagnoses were classified as having suspected cases of herpes zoster infection (1).

3. Herpes zoster ophthalmicus: Subjects with herpes zoster infection in the ocular region (1). It is the reactivation of latent varicella-zoster virus (VZV) in the first and/or second division of the trigeminal nerve (2)

4. Post-herpetic neuralgia: Subjects with pain continuing 90 days or longer after the onset of the shingles rash. Effectiveness data are based on this definition and also known as PHN-90 (1). Any pain persisting after rash healing to pain that has persisted at least 60, 90, 120 days or 6 months after rash onset to clinically meaningful pain 3 months or more after rash onset (2).

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5. Quality-of-life: Measured by EuroQol (EQ5D), Health Utilities Index Mark 2 (HUI2), Health Utilities Index Mark 3 (HUI3) or Short Form 6 Dimensions (SF6D) (1).

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ABSTRACT

Introduction

Herpes zoster (HZ) and associated complications inflict substantial morbidity, and associated healthcare and socioeconomic burdens. Current treatments are not fully effective, especially among the most vulnerable populations. Two HZ vaccines are available and are part of the national immunisation programmes in many countries. This review will evaluate the effectiveness of zoster vaccines against incident herpes zoster and postherpetic neuralgia in adults 50 years and older.

Methods and analysis

The key information sources that will be searched include Medline (Ovid), Embase (Ovid), Cochrane libraries, and CINAHL. This search will consider post-licensure observational studies published in all languages between 2006 and 2020 that assessed the effectiveness of herpes zoster/zoster vaccines in adults 50 years and older. The identification of studies will be complemented with the search of reference lists and citations, and contact with authors of papers to request missing or additional data, where required. Following the search, all identified citations will be collated, and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review. Selected studies will follow the process of critical appraisal, data extraction, and data synthesis. Statistical analyses will be performed using a random-effects model.

Ethics and dissemination

Formal ethical approval is not required, as primary data will not be collected. The review will be disseminated in peer-reviewed publications and conference presentations.

Keywords: Herpes zoster vaccine, Effectiveness

Strengths and limitations of this study

- Systematic review of high-quality observational studies.
- The selection of studies, data extraction, and a quality assessment will be conducted by two independent authors.
- The protocol has been created according to published PRISMA-P guidelines.
- We do not have any language limitations.

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- Observational studies may produce evidence of possible bias.

INTRODUCTION

Varicella-zoster virus (VZV) is the aetiological agent of varicella, a highly infectious, acute, self-limiting, viral disease with serious complications in neonates, pregnant women, and immunocompromised persons (1). It is an exclusively human neurotropic alpha herpes virus (2). After primary infection, it becomes latent in the cranial nerve ganglia, dorsal root ganglia, and autonomic ganglia along the entire neuroaxis (2). Administration of varicella vaccine or natural infection generates VZV-specific antibody and VZV-specific T cell-mediated immunity (2).

The memory cell response contributes to the protection following re-exposure to VZV (3). This immunity is subsequently boosted either by endogenous re-exposure (subclinical reactivation of latent VZV) or exogenous re-exposure (3). VZV-specific T cell-mediated immunity is necessary to maintain latent VZV in a subclinical state in the sensory ganglia (4). A decline in VZV-specific cell-mediated T cell-mediated immunity associated with ageing (immunosenescence), immune-suppressive diseases or therapies causes reactivation of latent VZV as herpes zoster (4) .

HZ presents as a unilateral vesicular rash, characteristically restricted to a single dermatome, and accompanied by radicular pain (5). It is estimated that each year there are one million cases in USA, 130,000 new cases in Canada, and more than 1.7 million cases in Europe (6-9). In New Zealand, more than 9,000 zoster hospitalisations have been recorded in the past 10 years (10). HZ incidence ranges from 3·4 to 5·0 per 1000 person-years, and 8 to 11 per 1000 person-years over the age of 65. One in four people is at risk of developing HZ during their lifetime, and two-thirds of people with the disease are aged 50 years or older (11). By the age of 85 years, >50% of the population report at least one episode of HZ (2). Major risk factors for HZ are increasing age, and a decline in cell-mediated immunity (5).

Laboratory confirmation of HZ occurs through the detection of VZV DNA using polymerase chain reaction or isolating VZV in cell culture from vesicular fluid, crusts, saliva, cerebrospinal fluid, or other specimens (5).

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The main complications of HZ include post-herpetic neuralgia (PHN - pain persisting more than 90 days after rash onset) in 22% (8-26%) of HZ patients; herpes zoster ophthalmicus (HZO), ophthalmic division of trigeminal nerve ~15% of cases (if untreated, 50-70% develop acute ocular complications, chronic complications, reduced vision, and blindness) (5).

The immune response can be boosted by the VZV vaccine developed to prevent HZ. Vaccination is, therefore, an important tool to reduce the epidemiological, clinical and economic burden of HZ, and to reduce the negative impact on the quality of life (2). Currently, two HZ vaccines are available. The recombinant zoster vaccine (RZV), approved in 2017, and the Zoster Vaccine Live (ZVL) licensed in 2006 (12). The effectiveness and public health impact of these vaccines have been demonstrated in many studies (13-19). The effectiveness of ZVL against an episode of HZ decreases gradually after vaccination, from 69% to 50% during the first year to about 17% by the seventh year after vaccination (20). RZV is also known as adjuvanted recombinant subunit zoster vaccine and consists of recombinant VZV glycoprotein E, and a liposome-based $AS01_B$ adjuvant system (12).

HZ and associated complications inflict substantial morbidity, and associated healthcare, and socioeconomic burdens (19). In the US, HZ and associated diseases result in about $1·3 billion in medical care costs and $1·7 billion in indirect costs annually, and this burden will increase in the coming years due to the ageing populations (19). Current treatments are not fully effective, especially among the most vulnerable patients (21). The best available option is prevention through vaccination. These vaccines have been licensed and marketed in many high-income countries (1, 22) like the United States, New Zealand, Australia, Germany, Canada, France, and Japan (Appendix I).

A 2019 Cochrane systematic review of 24 studies (randomised controlled trials or quasi-randomised controlled trials) involving 88,531 participants, aged 60 years and above showed that ZVL and RZV are effective in preventing herpes zoster disease for up to three years (23). This review excluded (a) trials involving immunosuppressed persons, (b) trials in people <60 years, and (c) observational studies.

In 2018, Tricco et al compared the efficacy, effectiveness, and safety of the herpes zoster live attenuated vaccine with the herpes zoster adjuvant recombinant subunit vaccine or placebo for adults aged 50 and older. They concluded that the RZV might prevent more cases of HZ than using the ZVL, but the RZV also carries a greater risk of adverse events at injection sites

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(11). Of all the 27 studies included, there were only four observational studies (1 case-control, and 3 cohort studies). For the laboratory or doctor confirmed cases of HZ, ZVL was not statistically different from placebo. There is a need to further explore these findings in observational studies.

Another systematic review estimated the relative efficacy and safety of vaccines for the prevention of HZ using network meta-analysis. RZV was significantly more effective in reducing HZ and PHN incidence in adults 60 years, and above, compared with ZVL. RZV resulted in more reactogenicity following immunization (24). The review excluded (a) trials conducted in immunocompromised patients, (b) trials in participants less than 50 years, and (c) observational studies.

In 2019, Senderovich et al, summarized the current literature available on the efficacy of the HZ vaccines in adults over 60 years old, and evaluated the cost-effectiveness of the HZ vaccines. 10 studies that met the inclusion criteria highlighted the efficacy of the HZ vaccines (6). The review focused on adults over 60 years old residing in long-term care facilities, and was limited to a five-year period (January 2013 to April 2018).

There is a need for a systematic review that will summarise real-world evidence (25) of the effectiveness of herpes zoster vaccines from post-licensure studies with different designs, conducted in different populations. Vaccine performance has been shown to vary in routine public health practice (under real world conditions) (26). Vaccine effectiveness (may be different from efficacy observed in trials) is influenced by host factors (age, comorbidity, previous exposure, co-administered vaccines and drugs and time since vaccination), vaccine characteristics (mode of administration, vaccine type and composition), and epidemiological factors (circulating strains, force of infection and herd immunity). Vaccine effectiveness (protection attributable to a vaccine administered under field conditions to a given population) measured by observational post-licensure studies (27) is important to inform public health programs and policies. The evidence will be obtained by searching, critically appraising, and synthesising the evidence from observational studies of published, and unpublished literature.

This present study will be the only review that will evaluate, and summarise the post-licensure effectiveness of HZ vaccines from 2006 to 2020 in adults 50 years and above.

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A preliminary search of PROSPERO, MEDLINE, the Cochrane Database of Systematic Reviews and the *JBI Database of Systematic Reviews and Implementation Reports* was conducted and no current or underway systematic reviews on the topic were identified.

OBJECTIVES

We seek to evaluate the effectiveness of zoster vaccines in adults 50 years and older. This includes incidence of herpes zoster, postherpetic neuralgia and herpes zoster ophthalmicus (primary outcomes), and herpes zoster related hospitalisations and quality of life (secondary outcomes)).

METHODS

The proposed systematic review will be conducted in accordance with the Joanna Briggs Institute methodology for systematic reviews of effectiveness evidence (28, 29).

Design

We will conduct a systematic review of published and unpublished studies.

Eligibility criteria

Types of studies: This review will consider analytical observational studies including prospective and retrospective cohort studies, and case-control studies. Studies published in all languages will be included. Studies published from 2006 to present will be included as the first herpes zoster vaccine was licensed in 2006. Systematic reviews and meta-analyses will be used only for identifying additional studies; these will not be included in this review.

Types of participants: The review will consider studies that include adults aged 50 years and older (studies can be included when adults younger than 50 are included, so long as this is not the majority).

Types of interventions: This review will consider studies that evaluate the effectiveness of herpes zoster vaccines. The intervention is being vaccinated (live attenuated injectable herpes zoster vaccine, adjuvant recombinant subunit herpes zoster vaccine). i.e. any licensed HZ vaccine (including all preparations, schedules, dosing, and routes of administration).

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Incomplete vaccination (failure to get the second dose of recombinant vaccine) will be considered unvaccinated.

Types of comparators: This review will consider studies that compare the intervention to no vaccination, placebo, or other vaccine

Types of outcome measures: This review will consider studies that include the following outcomes: The outcome measures are defined in Appendix II.

Primary outcomes

1. Incidence of herpes zoster (entire duration of follow-up or repeated measures).
   - Number of cases and person-years
   - Incidence rate per arm
2. Incidence of Postherpetic neuralgia (entire duration of follow-up)
   - 90 days
   - Number of cases and person-years
   - Incidence rate per arm
3. Incidence of other HZ complications (herpes zoster ophthalmicus)
   - Number/proportion of events or number/proportion of patients with the event
   - Incidence rate

Secondary outcomes

- Herpes zoster related hospitalisations
- Quality of life

These outcomes will be measured by vaccine effectiveness.

Types of studies

This review will consider analytical observational studies including prospective and historical cohort studies, and case-control studies. Studies published in all languages will be included. Studies published from 2006 to present will be included as the first herpes zoster vaccine was licensed in 2006.

Search Methods

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The search strategy will aim to locate both published and unpublished studies. An initial limited search of MEDLINE was undertaken to identify articles on the topic. The text words contained in the titles and abstracts of relevant articles, and the index terms used to describe the articles were used to develop a full search strategy for Medline (Ovid) (Appendix III). The search strategy, including all identified keywords and index terms, will be adapted for each included information source. The reference list of all studies selected for critical appraisal will be screened for additional studies.

The databases to be searched include Medline (Ovid), Embase (Ovid), Cochrane libraries, Cumulative Index to Nursing and Allied Health Literature (CINAHL), ProQuest Central, and Dimensions. Sources of unpublished studies and grey literature to be searched include CareSearch, Grey Literature Report, Google, and World Health Organisation (WHO).

Study selection

Following the search, all identified citations will be collated and uploaded into EndNote X9 (Clarivate Analytics, PA, USA), and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review using Covidence. Potentially relevant studies will be retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI) (Joanna Briggs Institute, Adelaide, Australia) (29).

The full text of selected citations will be assessed in detail against the inclusion criteria by two independent reviewers. Reasons for exclusion of full-text studies that do not meet the inclusion criteria will be recorded and reported in the systematic review. Any disagreements that arise between the reviewers at each stage of the study selection process will be resolved through discussion, or with a third reviewer. The results of the search will be reported in full in the final systematic review and presented in Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) flow diagram (30).

Assessment of methodological quality

Eligible studies will be critically appraised by two independent reviewers at the study level for methodological quality in the review using standardized critical appraisal instruments

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from the Joanna Briggs Institute for Cohort and case control studies (Appendix IV) (28, 29). This will help us assess how each study has addressed bias in its design, conduct and analysis. Authors of papers will be contacted to request missing or additional data for clarification, where required. Any disagreements that arise will be resolved through discussion, or with a third reviewer. The results of the critical appraisal will be reported in narrative form and in a table.

All studies, regardless of the results of their methodological quality, will undergo data extraction and synthesis (where possible).

Data extraction

Data will be extracted from studies included in the review by two independent reviewers using the standardised data extraction tool.

The data extracted will include specific details about the populations, study methods, interventions, and outcomes of significance to the review objective (HZ, HZ ophthalmicus, postherpetic neuralgia, HZ related hospitalisations, and quality of life).

Any disagreements that arise between the reviewers will be resolved through discussion, or with a third reviewer. Authors of papers will be contacted to request missing or additional data, where required.

Data analysis

Studies will, where possible be pooled in a statistical meta-analysis using JBI SUMARI (29). Effect sizes will be expressed as either odds ratios or relative risk (for dichotomous data) and weighted (or standardized) final post-intervention mean differences (for continuous data) and their 95% confidence intervals will be calculated.

Heterogeneity will be assessed statistically using the standard chi-squared and I squared tests. Statistical analyses will be performed using a random-effects model (31). We will be reviewing studies that compare the proportion of people developing zoster in vaccinated and unvaccinated groups. We expect the effect sizes (OR, RR etc) to be similar but not the same. The effect size may vary depending on age and mix of participants, health status, and programme implementation (32). Hence the random-effects model (rather than the fixed-

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effect model) is more appropriate. We will assess pooled vaccine effectiveness by vaccine, time since vaccination, disease (zoster, PHN, herpes zoster ophthalmicus etc), and by study design.

Sensitivity analyses will be conducted to test decisions made regarding the impact of excluding or including studies in a meta-analysis based on sample size, or methodological quality. If results remain consistent across the different analyses, the results can be considered robust as even with different decisions they remain the same/similar. If the results differ across sensitivity analyses, this is an indication that the result may need to be interpreted with caution (28).

Where statistical pooling is not possible the findings will be presented in narrative form including tables and figures to aid in data presentation where appropriate. A funnel plot will be generated to assess publication bias if there are 10 or more studies included in a meta-analysis. Statistical tests for funnel plot asymmetry (Egger test, Begg test, Harbord test) will be performed where appropriate.

Assessing certainty in the findings

The Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach for grading the certainty of evidence will be followed (28, 33), and a Summary of Findings (SoF) will be created using GRADEPro GDT (McMaster University, ON, Canada). The SoF will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence-based on the risk of bias, directness, heterogeneity, precision and risk of publication bias of the review results. The outcomes reported in the SoF will be herpes zoster, herpes zoster ophthalmicus, postherpetic neuralgia, herpes zoster related hospitalisations and quality of life.

Patient and Public Involvement statement

No patient involved

Ethics and dissemination

Formal ethical approval is not required, as primary data will not be collected. The findings will be disseminated in peer-reviewed journals and conference presentations.

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Conclusion

This systematic review will provide evidence regarding the effectiveness of herpes zoster vaccination in people 50 years of age and above. The evidence will be obtained by searching, critically appraising, and synthesising the evidence from observational studies of published and unpublished literature.

Registration and publishing statement

The systematic review protocol will be registered with the PROSPERO International Prospective Register of Systematic Reviews, and reported using Preferred Reporting Items for Systematic Reviews and Meta-analysis checklist to guide the reporting of the review.

Contributors

MFJ wrote this review. CRS and BPN contributed to conceiving this review, defining the study question, and planning the review, and commented critically on several drafts of the protocol. PAM and PEE contributed on a draft of this review with critical comments.

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Conflicts of interest

Professor Simpson reports grants from the UK National Institute for Health Research, UK Medical Research Council, the NZ Health Research Council and Ministry of Business Innovation and Employment. Dr. Nguyen reports grants from the NZ Health Research Council and NZ Ministry of Business Innovation and Employment.

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Provenance and peer review

Not commissioned; externally peer-reviewed.

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Supplementary File

Appendix I: Zoster Vaccination

Country	Recommendation	Vaccine type
New Zealand	Adults aged 65 years, catch-up 66–80 years (end by 31 December 2020)	ZVL (Zostavax®)
Australia	Adults aged 70 years, and Catch-up 71–79 years (five-year program) until 2021	ZVL (Zostavax®)
USA	Adults age 50 and older	RZV (Shingrix®)
Austria	Adults aged ≥50 years	RZV (Shingrix®)
Germany	Adults aged ≥60 years and specific groups	RVZ (Shingrix®)
Canada	Adults aged ≥50 years	RVZ (Shingrix®)
France	Adults aged 65–74 years, Catch-up 75–79	ZVL (Zostavax®)
Spain	Adults aged ≥60 years (for specific risk groups)	ZVL (Zostavax®)
Greece	Adults aged ≥60 years	ZVL (Zostavax®)
Italy	Adults aged ≥65 years	ZVL (Zostavax®)
United Kingdom	Adults aged 70–78 years	ZVL (Zostavax®)
United Arab Emirates	Adults aged >60 years	ZVL (Zostavax®)

Appendix II: Outcome Definitions

1. Confirmed herpes zoster infection: Those who underwent clinical diagnosis of herpes zoster infection through laboratory testing (e.g. polymerase-chain-reaction assay, virus culture) and/or examination by a physician were classified as having confirmed cases of herpes zoster infection (1).

2. Suspected herpes zoster infection: Subjects with non-injection-site herpes-zoster like rashes, varicella-like rashes, vesicular rashes or unilateral rashes (as defined by the investigators) and no other alternative clinical diagnoses were classified as having suspected cases of herpes zoster infection (1).

3. Herpes zoster ophthalmicus: Subjects with herpes zoster infection in the ocular region (1). It is the reactivation of latent varicella-zoster virus (VZV) in the first and/or second division of the trigeminal nerve (2)

4. Post-herpetic neuralgia: Subjects with pain continuing 90 days or longer after the onset of the shingles rash. Effectiveness data are based on this definition and also known as PHN-90

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(1). Any pain persisting after rash healing to pain that has persisted at least 60, 90, 120 days or 6 months after rash onset to clinically meaningful pain 3 months or more after rash onset (2).

5. Quality-of-life: Measured by EuroQol (EQ5D), Health Utilities Index Mark 2 (HUI2), Health Utilities Index Mark 3 (HUI3) or Short Form 6 Dimensions (SF6D) (1)

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Appendix III: Search strategy

Key concepts (alternative words)

1. Herpes zoster (zoster, shingles, varicella-zoster virus, human herpesvirus 3)

2. Postherpetic neuralgia (PHN)

3. Herpes zoster ophthalmicus (HZO)

4. Herpes zoster vaccine (zoster vaccine, shingles vaccine, zostavax, live attenuated zoster vaccine, zoster vaccine live (ZVL), shingrix, adjuvant recombinant subunit vaccine (RZV), varicella-zoster vaccine).

5. Effectiveness (effective, performance, impact, efficacy, effect, efficiency).

6. Other search terms to be considered: HZ related hospitalisations, quality of life, immune status, comorbidities.

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Search conducted on Medline (Ovid) show the following

#	Search term	Results
1	exp Herpes Zoster/	11747
2	Herpesvirus 3, Human/	7028
3	Varicellovirus/	285
4	shingles.tw.	1240
5	(varicella adj3 virus*).tw.	8299
6	zoster.tw.	17343
7	(hhv3 or hhv-3).tw.	16
8	shingles.ab,ti.	1240
9	zoster.ab,ti.	17343
10	(varicella adj4 virus*).ab,ti.	8370
11	varicellovirus/de	27
12	varicellovir*.ab,ti.	62
13	hhv3.ab,ti.	5
14	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13	22283
15	exp Vaccines/	225746
16	exp Immunization/	173130
17	Vaccination/	80276
18	(vaccin* or immuni* or inocul*).tw.	636177
19	(vaccin* or immuni* or inocul*).ab,ti.	636177
20	15 or 16 or 17 or 18 or 19	724546
21	14 and 21	4827
22	Herpes Zoster Vaccine/	790
23	((zoster or shingles) adj3 vaccin*).tw.	1264
24	zostavax.tw,nm.	132
25	((sub-unit or subunit) adj3 (vaccin* or zoster)).ab,ti.	5893
26	(Zostavax or Shingrix).ab,ti.	153
27	22 or 23 or 24 or 25 or 26	7342
28	21 or 27	10650
29	exp cohort analysis/	1978236
30	exp longitudinal study/	132972
31	exp prospective study/	534668
32	cohort*.tw.	573867
33	exp case control study/	1069768
34	(case* and control*).tw.	688836
35	29 or 30 or 31 or 32 or 33 or 34	2751394
36	28 and 35	989
37	limit 36 to yr="2006 -Current"	696
38	remove duplicates from 37	692

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Appendix IV: The JBI Critical Appraisal tools for use in JBI Systematic Reviews

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JBI Critical Appraisal Checklist for Cohort Studies

Reviewer_____Date_____

Author _____Year_____Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the two groups similar and recruited from the same population?	☐	☐	☐	☐
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were confounding factors identified?	☐	☐	☐	☐
5. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?	☐	☐	☐	☐
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	☐	☐	☐	☐
10. Were strategies to address incomplete follow up utilized?	☐	☐	☐	☐
11. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

_____

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JBI Critical Appraisal Checklist for Case Control Studies

Reviewer_____Date_____

Author _____Year_____Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the groups comparable other than the presence of disease in cases or the absence of disease in controls?	☐	☐	☐	☐
2. Were cases and controls matched appropriately?	☐	☐	☐	☐
3. Were the same criteria used for identification of cases and controls?	☐	☐	☐	☐
4. Was exposure measured in a standard, valid and reliable way?	☐	☐	☐	☐
5. Was exposure measured in the same way for cases and controls?	☐	☐	☐	☐
6. Were confounding factors identified?	☐	☐	☐	☐
7. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
8. Were outcomes assessed in a standard, valid and reliable way for cases and controls?	☐	☐	☐	☐
9. Was the exposure period of interest long enough to be meaningful?	☐	☐	☐	☐
10. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

_____

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BMJ Open

Post-Licensure Herpes Zoster Vaccines Effectiveness: Systematic Review Protocol

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Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Infectious diseases
Keywords:	Epidemiology < INFECTIOUS DISEASES, Public health < INFECTIOUS DISEASES, PUBLIC HEALTH

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Post-Licensure Herpes Zoster Vaccines Effectiveness: Systematic Review Protocol.

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ABSTRACT

Introduction

Herpes zoster (HZ) and associated complications inflict substantial morbidity, and associated healthcare and socioeconomic burdens. Current treatments are not fully effective, especially

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among the most vulnerable populations. Two HZ vaccines are available and are part of the national immunisation programmes in many countries. This review will evaluate the effectiveness of zoster vaccines against incident herpes zoster and postherpetic neuralgia in adults 50 years and older.

Methods and analysis

The key information sources that will be searched include Medline (Ovid), Embase (Ovid), Cochrane libraries, and CINAHL. This search will consider post-licensure observational studies published in all languages between 2006 and 2020 that assessed the effectiveness of herpes zoster/zoster vaccines in adults 50 years and older. The identification of studies will be complemented with the search of reference lists and citations, and contact with authors of papers to request missing or additional data, where required. Following the search, all identified citations will be collated, and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review. Selected studies will follow the process of critical appraisal, data extraction, and data synthesis. Statistical analyses will be performed using a random-effects model.

Ethics and dissemination

Formal ethical approval is not required, as primary data will not be collected. The review will be disseminated in peer-reviewed publications and conference presentations.

Keywords: Herpes zoster vaccine, Effectiveness

Strengths and limitations of this study

- Systematic review of high-quality observational studies.
- The selection of studies, data extraction, and a quality assessment will be conducted by two independent authors.
- The protocol has been created according to published PRISMA-P guidelines.
- We do not have any language limitations.
- Observational studies may produce evidence of possible bias and confounding.

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INTRODUCTION

Varicella-zoster virus (VZV) is the aetiological agent of varicella, a highly infectious, acute, self-limiting, viral disease with serious complications in neonates, pregnant women, and immunocompromised persons (1). It is an exclusively human neurotropic alpha herpes virus (2). After primary infection, it becomes latent in the cranial nerve ganglia, dorsal root ganglia, and autonomic ganglia along the entire neuroaxis (2). Administration of varicella vaccine or natural infection generates VZV-specific antibody and VZV-specific T cell-mediated immunity (2).

The memory cell response contributes to the protection following re-exposure to VZV (3). This immunity is subsequently boosted either by endogenous re-exposure (subclinical reactivation of latent VZV) or exogenous re-exposure (3). VZV-specific T cell-mediated immunity is necessary to maintain latent VZV in a subclinical state in the sensory ganglia (4). A decline in VZV-specific cell-mediated T cell-mediated immunity associated with ageing (immunosenescence), immune-suppressive diseases or therapies causes reactivation of latent VZV as herpes zoster (4) .

HZ presents as a unilateral vesicular rash, characteristically restricted to a single dermatome, and accompanied by radicular pain (5). It is estimated that each year there are one million cases in USA, 130,000 new cases in Canada, and more than 1.7 million cases in Europe (6-9). In New Zealand, more than 9,000 zoster hospitalisations have been recorded in the past 10 years (10). HZ incidence ranges from 3·4 to 5·0 per 1000 person-years, and 8 to 11 per 1000 person-years over the age of 65. One in four people is at risk of developing HZ during their lifetime, and two-thirds of people with the disease are aged 50 years or older (11). By the age of 85 years, >50% of the population report at least one episode of HZ (2). Major risk factors for HZ are increasing age, and a decline in cell-mediated immunity (5).

Laboratory confirmation of HZ occurs through the detection of VZV DNA using polymerase chain reaction or isolating VZV in cell culture from vesicular fluid, crusts, saliva, cerebrospinal fluid, or other specimens (5).

The main complications of HZ include post-herpetic neuralgia (PHN - pain persisting more than 90 days after rash onset) in 22% (8-26%) of HZ patients; herpes zoster ophthalmicus

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(HZO), ophthalmic division of trigeminal nerve ~15% of cases (if untreated, 50-70% develop acute ocular complications, chronic complications, reduced vision, and blindness) (5).

The immune response can be boosted by the VZV vaccine developed to prevent HZ. Vaccination is, therefore, an important tool to reduce the epidemiological, clinical, and economic burden of HZ, and to reduce the negative impact on the quality of life (2). Currently, two HZ vaccines are available. The recombinant zoster vaccine (RZV) approved in 2017, and the Zoster Vaccine Live (ZVL) licensed in 2006 (12). The effectiveness and public health impact of these vaccines have been demonstrated in many studies (13-19). The effectiveness of ZVL against an episode of HZ decreases gradually after vaccination, from 69% to 50% during the first year to about 17% by the seventh year after vaccination (20). RZV is also known as adjuvanted recombinant subunit zoster vaccine and consists of recombinant VZV glycoprotein E, and a liposome-based $AS01_B$ adjuvant system (12).

HZ and associated complications inflict substantial morbidity, and associated healthcare, and socioeconomic burdens (19). In the US, HZ and associated diseases result in about $1·3 billion in medical care costs and $1·7 billion in indirect costs annually, and this burden will increase in the coming years due to the ageing populations (19). Current treatments are not fully effective, especially among the most vulnerable patients (21). The best available option is prevention through vaccination. These vaccines have been licensed and marketed in many high-income countries (1, 22) like the United States, New Zealand, Australia, Germany, Canada, France, and Japan (Appendix I).

A 2019 Cochrane systematic review of 24 studies (randomised controlled trials or quasi-randomised controlled trials) involving 88,531 participants, aged 60 years and above showed that ZVL and RZV are effective in preventing herpes zoster disease for up to three years (23). This review excluded (a) trials involving immunosuppressed persons, (b) trials in people <60 years, and (c) observational studies.

In 2018, Tricco et al compared the efficacy, effectiveness, and safety of the herpes zoster live attenuated vaccine with the herpes zoster adjuvant recombinant subunit vaccine or placebo for adults aged 50 and older. They concluded that the RZV might prevent more cases of HZ than using the ZVL, but the RZV also carries a greater risk of adverse events at injection sites (11). Of all the 27 studies included, there were only four observational studies (1 case-control, and 3 cohort studies). For the laboratory or doctor confirmed cases of HZ, ZVL was

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not statistically different from placebo. There is a need to further explore these findings in observational studies.

Another systematic review estimated the relative efficacy and safety of vaccines for the prevention of HZ using network meta-analysis. RZV was significantly more effective in reducing HZ and PHN incidence in adults 60 years, and above, compared with ZVL. RZV resulted in more reactogenicity following immunization (24). The review excluded (a) trials conducted in immunocompromised patients, (b) trials in participants less than 50 years, and (c) observational studies.

In 2019, Senderovich et al, summarized the current literature available on the efficacy of the HZ vaccines in adults over 60 years old, and evaluated the cost-effectiveness of the HZ vaccines. 10 studies that met the inclusion criteria highlighted the efficacy of the HZ vaccines (6). The review focused on adults over 60 years old residing in long-term care facilities and was limited to a five-year period (January 2013 to April 2018).

There is a need for a systematic review that will summarise real-world evidence (25) of the effectiveness of herpes zoster vaccines from post-licensure studies with different designs, conducted in different populations. Vaccine performance has been shown to vary in routine public health practice (under real world conditions) (26). Vaccine effectiveness (may be different from efficacy observed in trials) is influenced by host factors (age, comorbidity, previous exposure, co-administered vaccines and drugs and time since vaccination), vaccine characteristics (mode of administration, vaccine type and composition), and epidemiological factors (circulating strains, force of infection and herd immunity). Vaccine effectiveness (protection attributable to a vaccine administered under field conditions to a given population) measured by observational post-licensure studies (27) is important to inform public health programs and policies. The evidence will be obtained by searching, critically appraising, and synthesising the evidence from observational studies of published, and unpublished literature.

This present study will be the only review that will evaluate, and summarise the post-licensure effectiveness of HZ vaccines from 2006 to 2020 in adults 50 years and above.

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A preliminary search of PROSPERO, MEDLINE, the Cochrane Database of Systematic Reviews and the *JBI Database of Systematic Reviews and Implementation Reports* was conducted and no current or underway systematic reviews on the topic were identified.

OBJECTIVES

We seek to evaluate the effectiveness of zoster vaccines in adults 50 years and older. This includes incidence of herpes zoster, postherpetic neuralgia and herpes zoster ophthalmicus (primary outcomes), and herpes zoster related hospitalisations and quality of life (secondary outcomes)).

METHODS

The proposed systematic review will be conducted in accordance with the Joanna Briggs Institute methodology for systematic reviews of effectiveness evidence (28, 29).

Design

We will conduct a systematic review of published and unpublished studies.

Eligibility criteria

Types of studies: This review will consider analytical observational studies including prospective and retrospective cohort studies, and case-control studies. Studies published in all languages will be included. Studies published from 2006 to present will be included as the first herpes zoster vaccine was licensed in 2006. Systematic reviews and meta-analyses will be used only for identifying additional studies; these will not be included in this review.

Types of participants: The review will consider studies that include adults aged 50 years and older (studies can be included when adults younger than 50 are included, so long as this is not the majority).

Types of interventions: This review will consider studies that evaluate the effectiveness of herpes zoster vaccines. The intervention is being vaccinated (live attenuated injectable herpes zoster vaccine, adjuvant recombinant subunit herpes zoster vaccine). i.e. any licensed HZ vaccine (including all preparations, schedules, dosing, and routes of administration).

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Incomplete vaccination (failure to get the second dose of recombinant vaccine) will be considered unvaccinated.

Types of comparators: This review will consider studies that compare the intervention to no vaccination, placebo, or other vaccine

Types of outcome measures: This review will consider studies that include the following outcomes: The outcome measures are defined in Appendix II.

Primary outcomes

1. Incidence of herpes zoster (entire duration of follow-up or repeated measures).
   - Number of cases and person-years
   - Incidence rate per arm
2. Incidence of Postherpetic neuralgia (entire duration of follow-up)
   - 90 days
   - Number of cases and person-years
   - Incidence rate per arm
3. Incidence of other HZ complications (herpes zoster ophthalmicus)
   - Number/proportion of events or number/proportion of patients with the event
   - Incidence rate

Secondary outcomes

- Herpes zoster related hospitalisations
- Quality of life

These outcomes will be measured by vaccine effectiveness.

Types of studies

This review will consider analytical observational studies including prospective and historical cohort studies, and case-control studies. Studies published in all languages will be included. Studies published from 2006 to present will be included as the first herpes zoster vaccine was licensed in 2006.

Search Methods

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The search strategy will aim to locate both published and unpublished studies. An initial limited search of MEDLINE was undertaken to identify articles on the topic. The text words contained in the titles and abstracts of relevant articles, and the index terms used to describe the articles were used to develop a full search strategy for Medline (Ovid) (Appendix III). The search strategy, including all identified keywords and index terms, will be adapted for each included information source. The reference list of all studies selected for critical appraisal will be screened for additional studies.

The databases to be searched include Medline (Ovid), Embase (Ovid), Cochrane libraries, Cumulative Index to Nursing and Allied Health Literature (CINAHL), ProQuest Central, and Dimensions. Sources of unpublished studies and grey literature to be searched include CareSearch, Grey Literature Report, Google, and World Health Organisation (WHO).

Study selection

Following the search, all identified citations will be collated and uploaded into EndNote X9 (Clarivate Analytics, PA, USA), and duplicates removed. Titles and abstracts will then be screened by two independent reviewers for assessment against the inclusion criteria for the review using Covidence. Potentially relevant studies will be retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI) (Joanna Briggs Institute, Adelaide, Australia) (29).

The full text of selected citations will be assessed in detail against the inclusion criteria by two independent reviewers. Reasons for exclusion of full-text studies that do not meet the inclusion criteria will be recorded and reported in the systematic review. Any disagreements that arise between the reviewers at each stage of the study selection process will be resolved through discussion, or with a third reviewer. The results of the search will be reported in full in the final systematic review and presented in Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) flow diagram (30).

Assessment of methodological quality

Eligible studies will be critically appraised by two independent reviewers at the study level for methodological quality in the review using standardized critical appraisal instruments

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from the Joanna Briggs Institute for Cohort and case control studies (Appendix IV) (28, 29). This will help us assess how each study has addressed bias in its design, conduct and analysis. Authors of papers will be contacted to request missing or additional data for clarification, where required. Any disagreements that arise will be resolved through discussion, or with a third reviewer. The results of the critical appraisal will be reported in narrative form and in a table.

Most observational studies are carried out using population-based databases that reflect daily medical practice. Potential confounding is common in studies of associations between vaccination and zoster using large databases (31). Predetermined data collection, absence of researcher control, completeness of recorded information, and misclassification can potentially affect the interpretation of any observed associations (31, 32).

All studies, regardless of the results of their methodological quality, will undergo data extraction and synthesis (where possible).

Data extraction

Data will be extracted from studies included in the review by two independent reviewers using the standardised data extraction tool.

The data extracted will include specific details about the populations, study methods, interventions, and outcomes of significance to the review objective (HZ, HZ ophthalmicus, postherpetic neuralgia, HZ related hospitalisations, and quality of life). Information on immunosuppression will also be extracted and stratified if appropriate.

Any disagreements that arise between the reviewers will be resolved through discussion, or with a third reviewer. Authors of papers will be contacted to request missing or additional data, where required.

Data analysis

Studies will, where possible be pooled in a statistical meta-analysis using JBI SUMARI (29). Effect sizes will be expressed as either odds ratios or relative risk (for dichotomous data) and weighted (or standardized) final post-intervention mean differences (for continuous data) and their 95% confidence intervals will be calculated.

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Heterogeneity will be assessed statistically using the standard chi-squared and I squared tests. Statistical analyses will be performed using a random-effects model (33). We will be reviewing studies that compare the proportion of people developing zoster in vaccinated and unvaccinated groups. We expect the effect sizes (OR, RR etc) to be similar but not the same. The effect size may vary depending on age and mix of participants, health status, and programme implementation (34). Hence the random-effects model (rather than the fixed-effect model) is more appropriate. We will assess pooled vaccine effectiveness by vaccine, time since vaccination, disease (zoster, PHN, herpes zoster ophthalmicus etc), immune status, and by study design.

Sensitivity analyses will be conducted to test decisions made regarding the impact of excluding or including studies in a meta-analysis based on sample size, or methodological quality. If results remain consistent across the different analyses, the results can be considered robust as even with different decisions they remain the same/similar. If the results differ across sensitivity analyses, this is an indication that the result may need to be interpreted with caution (28).

Where statistical pooling is not possible the findings will be presented in narrative form including tables and figures to aid in data presentation where appropriate. A funnel plot will be generated to assess publication bias if there are 10 or more studies included in a meta-analysis. Statistical tests for funnel plot asymmetry (Egger test, Begg test, Harbord test) will be performed where appropriate.

Assessing certainty in the findings

The Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach for grading the certainty of evidence will be followed (28, 35), and a Summary of Findings (SoF) will be created using GRADEPro GDT (McMaster University, ON, Canada). The SoF will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence-based on the risk of bias, directness, heterogeneity, precision, and risk of publication bias of the review results. The outcomes reported in the SoF will be herpes zoster, herpes zoster ophthalmicus, postherpetic neuralgia, herpes zoster related hospitalisations and quality of life.

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Patient and Public Involvement statement

No patient involved

Ethics and dissemination

Formal ethical approval is not required, as primary data will not be collected. The findings will be disseminated in peer-reviewed journals and conference presentations.

Registration and publishing statement

The systematic review protocol will be registered with the PROSPERO International Prospective Register of Systematic Reviews and reported using Preferred Reporting Items for Systematic Reviews and Meta-analysis checklist to guide the reporting of the review.

Contributors

MFJ wrote this review. CRS and BPN contributed to conceiving this review, defining the study question, and planning the review, and commented critically on several drafts of the protocol. PAM and PEE contributed on a draft of this review with critical comments.

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Conflicts of interest

Professor Simpson reports grants from the UK National Institute for Health Research, UK Medical Research Council, the NZ Health Research Council and Ministry of Business

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Provenance and peer review

Not commissioned; externally peer-reviewed.

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Supplementary File

Appendix I: Zoster Vaccination

Country	Recommendation	Vaccine type
New Zealand	Adults aged 65 years, catch-up 66–80 years (end by 31 December 2020)	ZVL (Zostavax®)
Australia	Adults aged 70 years, and Catch-up 71–79 years (five-year program) until 2021	ZVL (Zostavax®)
USA	Adults age 50 and older	RZV (Shingrix®)
Austria	Adults aged ≥50 years	RZV (Shingrix®)
Germany	Adults aged ≥60 years and specific groups	RVZ (Shingrix®)
Canada	Adults aged ≥50 years	RVZ (Shingrix®)
France	Adults aged 65–74 years, Catch-up 75–79	ZVL (Zostavax®)
Spain	Adults aged ≥60 years (for specific risk groups)	ZVL (Zostavax®)
Greece	Adults aged ≥60 years	ZVL (Zostavax®)
Italy	Adults aged ≥65 years	ZVL (Zostavax®)
United Kingdom	Adults aged 70–78 years	ZVL (Zostavax®)
United Arab Emirates	Adults aged >60 years	ZVL (Zostavax®)

Appendix II: Outcome Definitions

1. Confirmed herpes zoster infection: Those who underwent clinical diagnosis of herpes zoster infection through laboratory testing (e.g. polymerase-chain-reaction assay, virus culture) and/or examination by a physician were classified as having confirmed cases of herpes zoster infection (1).

2. Suspected herpes zoster infection: Subjects with non-injection-site herpes-zoster like rashes, varicella-like rashes, vesicular rashes or unilateral rashes (as defined by the investigators) and no other alternative clinical diagnoses were classified as having suspected cases of herpes zoster infection (1).

3. Herpes zoster ophthalmicus: Subjects with herpes zoster infection in the ocular region (1). It is the reactivation of latent varicella-zoster virus (VZV) in the first and/or second division of the trigeminal nerve (2)

4. Post-herpetic neuralgia: Subjects with pain continuing 90 days or longer after the onset of the shingles rash. Effectiveness data are based on this definition and also known as PHN-90

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(1). Any pain persisting after rash healing to pain that has persisted at least 60, 90, 120 days or 6 months after rash onset to clinically meaningful pain 3 months or more after rash onset (2).

5. Quality-of-life: Measured by EuroQol (EQ5D), Health Utilities Index Mark 2 (HUI2), Health Utilities Index Mark 3 (HUI3) or Short Form 6 Dimensions (SF6D) (1)

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Appendix III: Search strategy

Key concepts (alternative words)

1. Herpes zoster (zoster, shingles, varicella-zoster virus, human herpesvirus 3)

2. Postherpetic neuralgia (PHN)

3. Herpes zoster ophthalmicus (HZO)

4. Herpes zoster vaccine (zoster vaccine, shingles vaccine, zostavax, live attenuated zoster vaccine, zoster vaccine live (ZVL), shingrix, adjuvant recombinant subunit vaccine (RZV), varicella-zoster vaccine).

5. Effectiveness (effective, performance, impact, efficacy, effect, efficiency).

6. Other search terms to be considered: HZ related hospitalisations, quality of life, immune status, comorbidities.

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Search conducted on Medline (Ovid) show the following

#	Search term	Results
1	exp Herpes Zoster/	11747
2	Herpesvirus 3, Human/	7028
3	Varicellovirus/	285
4	shingles.tw.	1240
5	(varicella adj3 virus*).tw.	8299
6	zoster.tw.	17343
7	(hhv3 or hhv-3).tw.	16
8	shingles.ab,ti.	1240
9	zoster.ab,ti.	17343
10	(varicella adj4 virus*).ab,ti.	8370
11	varicellovirus/de	27
12	varicellovir*.ab,ti.	62
13	hhv3.ab,ti.	5
14	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13	22283
15	exp Vaccines/	225746
16	exp Immunization/	173130
17	Vaccination/	80276
18	(vaccin* or immuni* or inocul*).tw.	636177
19	(vaccin* or immuni* or inocul*).ab,ti.	636177
20	15 or 16 or 17 or 18 or 19	724546
21	14 and 21	4827
22	Herpes Zoster Vaccine/	790
23	((zoster or shingles) adj3 vaccin*).tw.	1264
24	zostavax.tw,nm.	132
25	((sub-unit or subunit) adj3 (vaccin* or zoster)).ab,ti.	5893
26	(Zostavax or Shingrix).ab,ti.	153
27	22 or 23 or 24 or 25 or 26	7342
28	21 or 27	10650
29	exp cohort analysis/	1978236
30	exp longitudinal study/	132972
31	exp prospective study/	534668
32	cohort*.tw.	573867
33	exp case control study/	1069768
34	(case* and control*).tw.	688836
35	29 or 30 or 31 or 32 or 33 or 34	2751394
36	28 and 35	989
37	limit 36 to yr="2006 -Current"	696
38	remove duplicates from 37	692

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Appendix IV: The JBI Critical Appraisal tools for use in JBI Systematic Reviews

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JBI Critical Appraisal Checklist for Cohort Studies

Reviewer_____Date_____

Author _____Year_____Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the two groups similar and recruited from the same population?	☐	☐	☐	☐
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were confounding factors identified?	☐	☐	☐	☐
5. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?	☐	☐	☐	☐
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	☐	☐	☐	☐
10. Were strategies to address incomplete follow up utilized?	☐	☐	☐	☐
11. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

_____

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JBI Critical Appraisal Checklist for Case Control Studies

Reviewer_____Date_____

Author _____Year_____Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the groups comparable other than the presence of disease in cases or the absence of disease in controls?	☐	☐	☐	☐
2. Were cases and controls matched appropriately?	☐	☐	☐	☐
3. Were the same criteria used for identification of cases and controls?	☐	☐	☐	☐
4. Was exposure measured in a standard, valid and reliable way?	☐	☐	☐	☐
5. Was exposure measured in the same way for cases and controls?	☐	☐	☐	☐
6. Were confounding factors identified?	☐	☐	☐	☐
7. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
8. Were outcomes assessed in a standard, valid and reliable way for cases and controls?	☐	☐	☐	☐
9. Was the exposure period of interest long enough to be meaningful?	☐	☐	☐	☐
10. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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